CO3 AT3 – Data Interpretation

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1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

About the Question

The question is about an RL agent whose reward increases from 10 to 45 over multiple training episodes. We have to understand whether this increase shows improvement in the agent's learning and policy.

How It Works

The agent interacts with the environment and performs actions. After each action, it receives a reward. It uses this experience to improve its future decisions.

Data Analysis

Initially, the reward is only 10, so the agent is not performing very well. As training continues, the reward gradually increases and reaches 45. This shows that the agent is choosing better actions as it gains more experience.

RL Concept / Application

Reward is used to measure the performance of an RL agent. An increasing reward generally means the learned policy is becoming better.

Conclusion

The increase from 10 to 45 shows that the agent is learning successfully. Its policy is improving and moving towards a better or optimal policy.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

About the Question

The question compares an epsilon-greedy agent using □ = 0.1, 0.3 and 0.5. The highest cumulative reward is obtained at 0.1, while the lowest is obtained at 0.5.

How It Works

Epsilon controls the amount of exploration. A small epsilon means the agent mostly selects the action it already knows is good. A larger epsilon makes the agent try more new actions.

Data Analysis

With □ = 0.1, the agent explores less and mainly uses its best-known actions, so it receives the highest reward. At □ = 0.5, it explores much more and may choose actions that give lower rewards.

RL Concept / Application

This is the exploration vs exploitation concept. Exploration helps discover new actions, while exploitation uses the knowledge already learned.

Conclusion

According to the given data, □ = 0.1 performs best because the agent gets the highest cumulative reward with less exploration and more exploitation.

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

About the Question

The question compares Q-Learning and SARSA using their average rewards at 100, 200 and 300 episodes. Q-Learning has slightly higher rewards than SARSA.

How It Works

Both algorithms learn the value of actions and improve their policy through experience. Q-Learning learns toward the best possible next action, while SARSA learns using the action actually selected by the agent.

Data Analysis

At all three episode levels, Q-Learning gives slightly higher rewards. This means that Q-Learning is performing better in the given situation.

RL Concept / Application

The reward values can be used to compare the learning performance of the two algorithms. Consistently higher rewards indicate better performance for the observed data.

Conclusion

Based on the given results, Q-Learning performs better and appears to converge faster than SARSA in this particular case.

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance.

About the Question

The question studies how different discount factors affect the average reward. The average reward increases as the discount factor increases.

How It Works

The discount factor □ decides how much importance the agent gives to future rewards. A low value focuses more on immediate rewards, while a high value considers future rewards more strongly.

Data Analysis

At □ = 0.2, the agent gives less importance to future rewards. At 0.5, it considers them more. At □ = 0.9, future rewards receive much greater importance, and the average reward is also higher.

RL Concept / Application

The discount factor is important when the agent needs to make decisions that provide benefits in the future rather than only immediately.

Conclusion

□ = 0.9 is the best value among the given options because it produces the highest average reward and gives greater importance to long-term rewards.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.

About the Question

The question gives the success rate of an RL agent at different training stages. It increases from 45% at 50 episodes to 92% at 200 episodes.

How It Works

Success rate shows how often the agent successfully completes its task. Training allows the agent to learn from previous experiences and improve its decisions.

Data Analysis

At 50 episodes, the success rate is only 45%, showing that the agent is still making many mistakes. At 200 episodes, it reaches 92%, which is a major improvement.

RL Concept / Application

An increasing success rate indicates that the agent's policy is becoming more effective. More training has helped it choose better actions.

Conclusion

The increase from 45% to 92% shows strong learning and policy improvement. The agent appears to be close to an optimal policy, although 92% alone does not prove that it is perfectly optimal

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability.

About the Question

The question compares learning rates 0.01, 0.05 and 0.1 in the CartPole environment. The agent has better stability with 0.05.

How It Works

Learning rate controls how much the agent changes its learned values after receiving new information. A very small rate makes learning slower, while a large rate makes bigger changes.

Data Analysis

With 0.01, learning may be slow because the updates are small. With 0.1, the updates are larger and can cause instability. The value 0.05 provides a better balance.

RL Concept / Application

Choosing a suitable learning rate is important for achieving both good learning speed and stable convergence.

Conclusion

Based on the given observation, 0.05 is the best learning rate because it provides a good balance between speed and stability.

7. A Q-learning agent is tested with different exploration strategies: ε-greedy, softmax, and random exploration. The cumulative reward is highest with ε-greedy. Compare the strategies and defend why ε-greedy achieves better performance in this scenario.

About the Question

The question compares ε-greedy, softmax and random exploration. The cumulative reward is highest with ε-greedy.

How It Works

ε-greedy usually selects the best-known action but sometimes explores another action. Softmax gives actions different probabilities based on their values. Random exploration selects actions randomly.

Data Analysis

The highest cumulative reward is obtained using ε-greedy. This means that its way of balancing exploration and exploitation is more suitable for this environment.

RL Concept / Application

Exploration is needed to discover useful actions, while exploitation allows the agent to use actions that are already known to work well.

Conclusion

ε-greedy performs better in the given scenario because it provides a useful balance between exploration and exploitation and produces the highest reward.

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy

Initially, the agent takes 50 steps, showing that it does not know an efficient path. After training, the number falls to 15 steps. This is a reduction of 35 steps, showing a significant improvement.

RL Concept / Application

A reduction in the number of steps indicates that the agent has learned a more efficient policy for reaching the goal.

Conclusion

The decrease from 50 to 15 steps shows strong policy improvement and suggests that the agent is moving towards convergence. To confirm that the policy is fully optimal, we would need to know the shortest possible path.

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.

About the Question

The question compares SARSA using discount factors 0.4, 0.7 and 0.95. The highest long-term reward is obtained at 0.95.

How It Works

The discount factor determines how much importance the agent gives to future rewards. A higher value means that future rewards are considered more important.

Data Analysis

At 0.4, the agent focuses more on immediate rewards. At 0.7, it gives more consideration to future rewards. At 0.95, it strongly considers future rewards and produces the highest long-term reward.

RL Concept / Application

A high discount factor is useful when the agent needs to plan ahead and obtain rewards over a longer period.

Conclusion

$\gamma = 0.95$ is the most effective value in this case because it gives the highest long-term reward and supports future-oriented decision making.

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.

About the Question

The question describes an agent in a stochastic environment. Its rewards fluctuate during the early episodes but become stable after 500 episodes.

How It Works

The agent learns by trying different actions and observing their results. In a stochastic environment, outcomes can vary, so rewards may not be constant during the early learning stage.

Data Analysis

At first, the reward fluctuates because the agent is still exploring and learning. After about 500 episodes, the reward becomes more stable. This shows that the agent's decisions are becoming more consistent.

RL Concept / Application

Stable rewards indicate that the agent has learned a more consistent policy. It also suggests that the policy can handle the randomness of the environment better.

Conclusion

The stabilization of rewards after 500 episodes indicates policy improvement, better convergence and greater robustness of the agent's learning.